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Executive Summary of the Thesis

Parametric correlated noise modeling and filtering for restoration of stripe artifacts in electron microscopy

Laurea Magistrale in Computer Science and Engineering - Ingegneria Informatica

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1. Introduction

Electron Microscopy (EM) enables high-resolution imaging of biological samples by means of electron beams, and variations like Serial Block-Face Scanning Electron Microscopy (SBF-SEM), Focused Ion Beam Scanning Electron Microscopy (FIB-SEM), and Transmission Electron Microscopy (TEM) allow the digital three-dimensional reconstruction of specimens based on two-dimensional sequential scans. These EM variations require sample preparation by milling or mechanical sectioning, destructive processes that can introduce undesirable elongated deformations and grooves, commonly referred to as cutting artifacts, curtaining, or stripe artifacts [3, 6–8]. These artifacts degrade the visual appeal of the affected images and can impair image analysis workflows, e.g., identification and segmentation of biomedical structures and the three-dimensional reconstruction of specimens. Typical state-of-the-art methods that aim to restore stripe deformations are based on Fourier masking, filtering, and recovery, as well as on variational methods that optimize targeted objective functions. Instead, this work investigates whether the stripe artifacts can be effectively modeled as stationary spatially correlated noise and subsequently removed from EM images using non-blind denoising algorithms such as Block-Matching and 3D Filtering (BM3D) [1]. Therefore, the main contributions of the thesis are (i) a parametric model of noise correlation capable of representing stripe artifacts with various directions, intensities, periodicities, and uniformities, and (ii) an automatic denoising pipeline that identifies stripe noise in degraded images, estimates the parameters that optimally characterize it using the introduced parametric model, and removes it.
Results demonstrate that the proposed parametric model and pipeline efficiently and effectively restore both simulated and real electron microscopy acquisitions degraded with stripe artifacts. Objective quality metrics and visual assessment demonstrate the high performance of the developed method against other stripe removal approaches.

2. Problem formulation

Spatially correlated noise can describe multiple forms of dependencies across different spatial locations in a signal. It fits naturally in the presented research problem, assuming stripes phys-

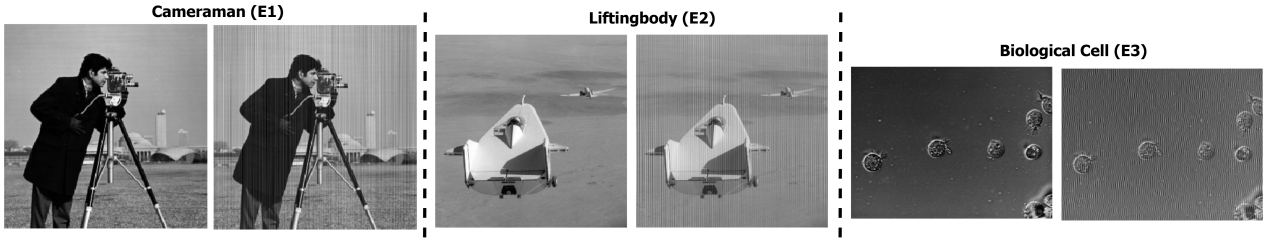


Figure 1: Examples of stripe artifacts addressed by this work. Left: Noise-free demo images from MATLAB. Right: Degraded images with added correlated stripe noise following (1) and (2).

ically affecting the specimens can be interpreted as a correlation of pixel values in the corresponding digital image.

Let Z be any noisy microscope image in grayscale modeled as a 2D signal function $Z : X \to \mathbb{R}$, defined over a finite two-dimensional domain $X \subset \mathbb{Z}^2$ such that $\mathrm{x} \in X$ is the variable representing the vectorized coordinate position of each image pixel. Then, the stationary additive correlated noise model is

$$\begin{aligned} Z(\mathrm{x}) &= Y(\mathrm{x}) + \eta(\mathrm{x}), \\ \eta &= \nu \circledast g, \qquad \nu(\cdot) \sim \mathcal{N}(0,1), \end{aligned} \tag{1}$$

where $\circledast$ stands for the convolution operator, Y is the deterministic noise-free value of the image, η represents the additive noise, and g is the correlation kernel that specifies the type of spatial correlation that the noise term displays. This work considers that whereas Z is known and degraded with stripe artifacts, Y and η are unknown, but the latter describes the stripes. Under this reasoning, the goal of the thesis is to propose a model for η and a denoising pipeline that computes a reliable stripe/noise-free estimate $\widehat{Y}$.

3. Related work

Stripe denoising algorithms are designed to remove elongated artifacts from stripe-degraded images, preserving non-stripe content. According to [8], in the context of EM images, these algorithms can be broadly categorized into Variational and Fourier-based methods.

Variational methods formulate stripe removal as an optimization problem that finds a stripe-free estimate of a degraded image by minimizing a constrained objective function. For instance, the Variational Stationary Noise Remover (VSNR) [3] models stripes as stationary noise consisting of manually defined elementary patterns, which often limits its adaptability to variations in stripe direction, frequency, and uniformity. Alternatively, the General Stripe Removal (GSR) [8] uses a constrained formulation based on Total Variation, gradients, and sparse representations, but its original implementation is limited to removing vertical stripes and may introduce non-stripe artifacts.

The Fourier-based methods perform stripe removal by exploiting the characteristics of the stripe structures in the frequency domain. The Wavelet Fourier-Filtering (WFF) [7] uses wavelet and Fourier transforms to identify and attenuate frequency bands of stripe content, and its effectiveness depends on the wavelet selection and stripe characteristics. Instead, the Multidirectional Stripe Remover (MDSR) [6] uses a non-subsampled contourlet transform to perform Fourier filtering along different directions, often giving better results, but with the risk of introducing non-stripe artifacts and blurring in the resulting stripe-free image.

These limitations motivate the development of a stripe denoising method with an easily interpretable and flexible modeling, as the one developed in this work.

4. Proposed Pipeline

For modeling η in (1), this work assumes the correlation kernel g reproduces the appearance of the stripe artifacts observed in noisy EM images by taking the shape of the real part of a Gabor filter. This shape can be described by a reduced set of parameters as masked sinusoidal fringes that visually simulate stripe artifacts with varying characteristics such as direction, periodicity, intensity, and uniformity. Despite admitting a simple expression, the parameter interpretation and estimation are challenging.

To simplify the analysis, it is proposed to charac-

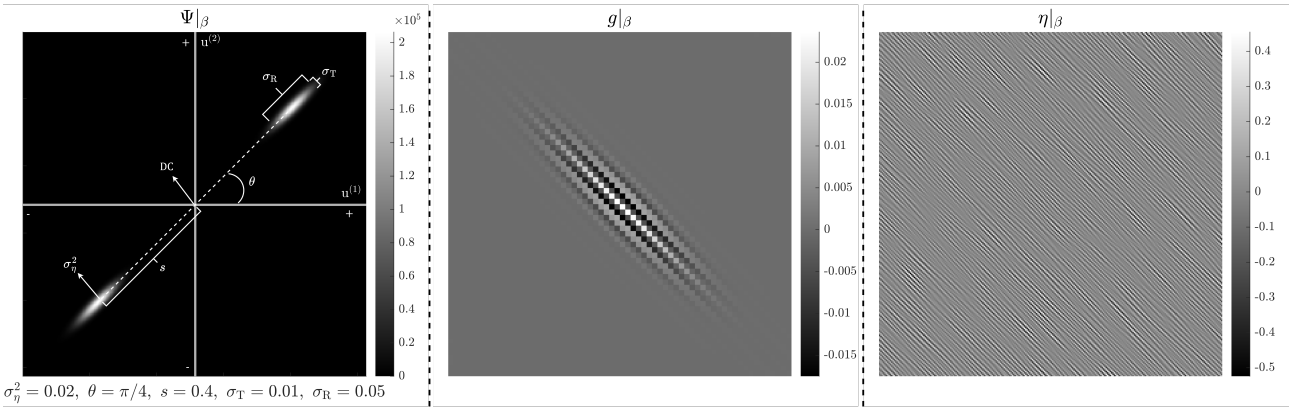


Figure 2: From left to right: a PSD term $\Psi|_\beta$ of 256×256 following (2), and a correlation kernel $g|_\beta$ of 64×64 with a noise η_β of 256×256 associated with the left PSD. Note that $g|_\beta$ and η_β displays presence of stripe patterns characterized by the parameters $s,\ \theta,\ \sigma_R,\ \sigma_T,\ \sigma_\eta^2$ of $\Psi|_\beta$.

terize stripe artifacts in the frequency domain by obtaining an equivalent expression to the Gabor g, but for the noise power spectral density function (PSD), which is defined as $\Psi := \text{var}\{\mathcal{F}[\eta]\}$ and which is related to the correlation kernel g as $\Psi = |X||\mathcal{F}[g]|^2$, where $\mathcal{F}[g]$ denotes the 2D Fourier transform of g and $|X|$ is the cardinality of the domain used as a scaling factor.
The PSD associated with the proposed expression for g is

$$\Psi|_\beta = \frac{\sigma_\eta^2}{|X|^{-2}\ \|\Gamma\|_1}\Gamma\ , \tag{2}$$

where Γ is the superposition of Gaussian pulses

$$\begin{aligned}\Gamma =& e^{-\frac{1}{2}\left((\mathrm{u}_\theta-\mathrm{S})^T\ \widetilde{\Sigma}^{-1}\ (\mathrm{u}_\theta-\mathrm{S})\right)} + \\ & e^{-\frac{1}{2}\left((\mathrm{u}_\theta+\mathrm{S})^T\ \widetilde{\Sigma}^{-1}\ (\mathrm{u}_\theta+\mathrm{S})\right)} + \\ & 2e^{-\frac{s^2}{2\sigma_R^2}} e^{-\frac{1}{2}\left(\mathrm{u}_\theta^T\ \widetilde{\Sigma}^{-1}\ \mathrm{u}_\theta\right)}\ ,\end{aligned} \tag{3}$$
$$\mathrm{S} = \begin{bmatrix} s \\ 1 \end{bmatrix}, \qquad \widetilde{\Sigma} = \begin{bmatrix} \sigma_R^2 & 0 \\ 0 & \sigma_T^2 \end{bmatrix}\ .$$

Here, $\mathrm{u}_\theta = R(\theta)\mathrm{u}$ denotes an anti-clockwise rotated coordinate frequency system, the T superscript denotes the transpose operation, and β is the vector of parameters that parametrize (2) defined as

$$\begin{aligned}\beta &:= \left[\sigma_\eta^2,\ \theta,\ s,\ \sigma_T,\ \sigma_R\right]\ , \\ \sigma_\eta^2 &\in \mathbb{R}_{\geq 0}\ , \theta \in [-90°, 90°)\ , \\ s,\ \sigma_R,\ \sigma_T &\in \mathbb{R}_{>0}\ .\end{aligned} \tag{4}$$

Figure 2 exemplifies a parametric PSD following (2), illustrating that this term consists of two dominant Gaussian pulses with the same spread and symmetrically centered at $\pm S$ in the Fourier domain. A third pulse centered at the zero-frequency component (DC) is also considered by (2), but because its amplitude decreases exponentially with $\frac{s^2}{2\sigma_R^2}$, its influence is often negligible, as in the term $\Psi|_\beta$ of Figure 2, where the centered pulse is not visible due to a sufficiently high s value.
The PSD in Figure 2 also illustrates the interpretation of each quantity parameterizing (2): While s and θ represent the shift and orientation of the pulses, σ_R and σ_T represent their spreads in the radial and tangential directions. Moreover, the parametric PSD in (2) is such that

$$\sigma_\eta^2 = |X|^{-2}\|\Psi|_\beta\|_1\ , \tag{5}$$

and consequently, the parameter σ_η^2 represents the variance of the underlying stripe noise η, controlling the height and energy of the pulses in $\Psi|_\beta$. On the whole, these parameters modify different visual aspects of the stripe artifact representation, as in Figure 1, where stripes parameterized by different values display modified appearances. Indeed, if an estimate of the noise PSD is obtained from a stripe-degraded image, like those in Figure 1, the parameters that characterize the artifacts can be estimated from this term, effectively modeling the stripes as correlated noise. This is the idea of the denoising pipeline detailed next.

4.1. Image detrending

As illustrated in Figure 3, the pipeline starts by detrending the input degraded image Z to iden-

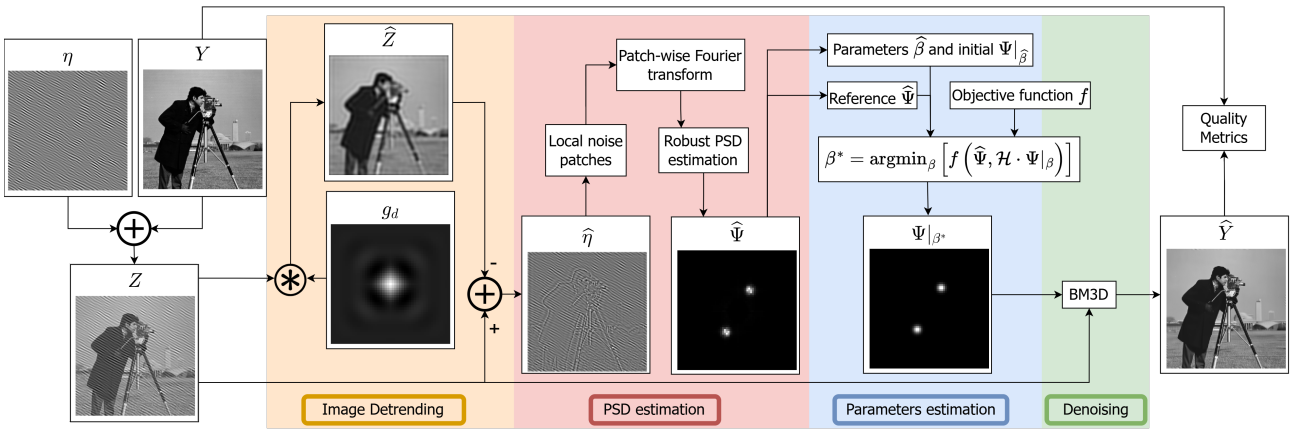


Figure 3: Denoising pipeline proposed in this work to remove stripe artifacts. Each color represents a different stage of the pipeline, starting with the input degraded image Z and ending with the output stripe-free estimate $\widehat{Y}$. The uncolored parts show the image formation of Z and the denoising assessment using a quality metric.

tify stripe content. A smoother version of Z is produced as $\widehat{Z} = g_d \circledast Z$, where g_d is a detrending kernel equivalent to a Local Polynomial Approximation estimator [4].

Thereafter, the detrending produces a residual $\widehat{\eta} = Z - \widehat{Z}$. Assuming natural non-stripe image content is predominantly characterized by low-frequency patterns like the ones seen in $\widehat{Z}$, the term $\widehat{\eta}$ is predominantly characterized by high-frequency stripe content.

4.2. PSD estimation

The residual $\widehat{\eta}$ is a coarse approximation of the stripe noise η that may include non-stripe high-frequency image content and consists of a single realization, so the associated PSD cannot be directly computed from its definition. Nonetheless, by exploiting the stationarity assumption of the noise model, from $\widehat{\eta}$ can be extracted noise patches to be treated as individual realizations of the same stationary noise process. Their Fourier transforms can then be used to estimate the variability of each frequency coefficient robustly, computing the PSD estimate $\widehat{\Psi} \in \mathbb{R}^{n \times m}_{\geq 0}$ as proposed in [2], where $n \times m$ represents the size of the extracted patches, smaller than the size of $\widehat{\eta}$ and Z.

Note that $\widehat{\Psi}$ does not directly approximate the corresponding ground-truth PSD Ψ, but approximates a high-pass filtered version of Ψ represented over a smaller frequency domain. Specifically, $\widehat{\Psi} \approx \mathcal{H} \cdot \Psi$, where $\mathcal{H} := |\mathcal{F}[\delta_0 - g_d]|^2$ represents a high-pass filter applied over Ψ and δ_0 is a centered Dirac delta function with mass 1.

4.3. Parameters estimations

Although $\widehat{\Psi}$ is not a direct estimate of the underlying noise PSD responsible for the stripe artifacts, since it is a high-pass filtered estimate with potential traces of non-striped content, we still assume $\widehat{\Psi}$ possesses dominant pulse structures associated with the characterization of stripes that follow the parametric PSD expression (2). So, considering the interpretation of each parameter illustrated in Figure 2 and explained after its definition in (4), the shift, orientation, energy, and radial and tangential spreads of the dominant pulses in $\widehat{\Psi}$ are estimated to build the parameter vector $\widehat{\beta}$ that approximates the stripe characterization.

The estimated parameters included in $\widehat{\beta}$ are then refined to overcome any potential non-stripe content leaked into the PSD estimate $\widehat{\Psi}$ that may alter the values in $\widehat{\beta}$. Hence, an optimization procedure is proposed to find the parameters β^* that characterize a PSD with expression (2) that closely resembles the dominant pulse structures of $\widehat{\Psi}$. Formally, the problem is

$$\beta^* = \arg\min_{\beta} \left[f\left(\widehat{\Psi}\ ,\ \mathcal{H} \cdot \Psi|_\beta\right)\right], \qquad (6)$$

where $\widehat{\Psi}$ is the reference PSD, $\Psi|_\beta$ denotes the fitted PSDs parameterized by some β parameters, and $\mathcal{H} \cdot \Psi|_\beta$ is the high-pass filtered version of the fitted PSD to be compared with $\widehat{\Psi}$. Additionally, f is an objective function used to mea-

sure the dissimilarities between the PSDs, and β^* are the optimal parameters that effectively minimize f.
The proposed objective function is

$$f\left(\widehat{\Psi}, \mathcal{H}\cdot\Psi|_\beta\right) \coloneqq \lambda \left\|[R]_+\right\|_F + \left\|[-R]_+\right\|_F, \tag{7}$$

where $R = \widehat{\Psi} - \mathcal{H}\cdot\Psi|_\beta$ denotes the residuals between the reference and the fitting PSDs, $[\,]_+$ represents an element-wise maximum operation that keeps only strictly positive values, $\|\ \|_F$ denotes the Frobenius norm, and $\lambda \in \mathbb{R}_{\geq 0}$ is the weighting factor for the positive residuals of R. This objective function allows assigning a different weight to the positive and negative residuals of R, letting the optimization problem adapt to different circumstances where the estimate $\widehat{\Psi}$ is influenced by non-stripe content.
To solve this optimization problem, the Nelder-Mead simplex algorithm [5] is used. This algorithm requires a proper initial guess for the optimized parameters to improve global convergence. For this reason, the parameters $\widehat{\beta}$ are used to build an initial parametric guess $\Psi|_{\widehat{\beta}}$ with which to compare to $\widehat{\Psi}$.

4.4. Denoising

The obtained β^* is used to generate a final parametric PSD $\Psi|_{\beta^*}$ following (2). This PSD follows the dominant pulse structures in $\widehat{\Psi}$ with reduced influence of non-striped content, and therefore models the statistical properties of the underlying stripe noise affecting Z.
The PSD $\Psi|_{\beta^*}$ can be supplied to different non-blind denoising algorithms to perform stripe noise removal, defining its output simply as

$$\widehat{Y} = \mathcal{D}\left(Z, \Psi|_{\beta^*}\right), \tag{8}$$

where $\mathcal{D}$ refers to the chosen denoiser method and $\widehat{Y}$ is the stripe-free estimate of Y in which the presence of artifacts should ideally be reduced compared with the degraded image Z.
The proposed pipeline is independent of the specific non-blind denoiser used in this last stage, but BM3D is appropriate due to its widespread use and performance. Moreover, BM3D can use the input PSD to adjust its collaborative filtering to the type of correlation represented by the PSD, which commonly leads to better noise filtering results.

5. Experiments and discussion

To demonstrate the effectiveness of the proposed pipeline for stripe removal, the synthetically degraded stripe images displayed in Figure 1 are used. Each image can then be passed to a denoising method as an input Z to obtain and estimate $\widehat{Y}$, with which to compute the Peak Signal-to-Noise Ratio (PSNR) metric with respect to the known noise-free term Y (also visible in Figure 1), assessing the reconstruction quality of the denoiser.

	Prop	GSR	MDSR	WFF	VSNR
E1	35.11	31.59	28.05	29.29	26.15
E2	43.18	36.28	37.90	31.38	27.67
E3	42.51	35.05	33.79	32.65	28.97

Table 1: Quality metrics (PSNR in dBs) of each considered stripe removal algorithm for the synthetic experiments E1, E2, and E3 of Figure 1.

Table 1 shows that the proposed pipeline (denoted as "Prop") achieves the highest PSNR in all the synthetic experiments compared to the methods GSR [8], MDSR [6], WFF [7], and VSNR [3]. While WFF and VSNR perform less consistently because they depend on case-specific parameters, GSR and MDSR show better PSNR results but still suffer from added blurring and non-stripe artifacts during denoising.
The pipeline is further tested on real SBF-SEM images from a dataset provided by I. Belevich (EMBI, University of Helsinki). Figure 4 shows the denoising of two markedly different samples affected by stripe noise, where the proposed pipeline successfully removes very subtle stripe artifacts while preserving other image content, as shown in the magnified crops.

6. Conclusion

This thesis demonstrates that stripe artifacts in EM images can be modeled as stationary spatially correlated noise with a parametric Gaussian PSD expression. The developed pipeline successfully identifies stripe noise in degraded images and computes a noise PSD estimate that non-blind denoisers can use to improve image quality. Synthetic experiments show the proposed pipeline's benefits over other state-of-the-

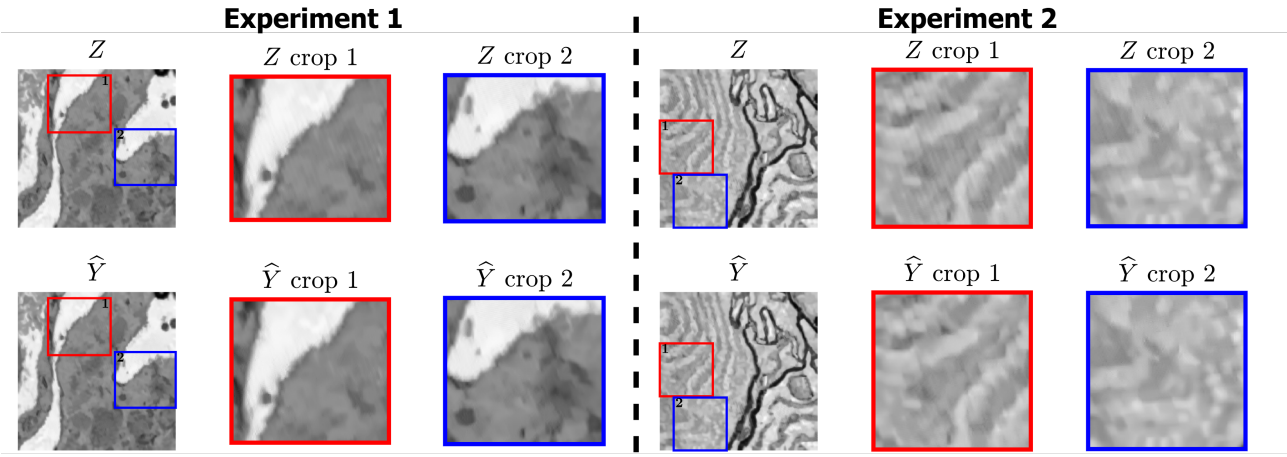


Figure 4: Experimental results of the proposed denoising pipeline over two real EM-degraded images. On the left of each experiment are both the corrupted image Z and the estimated noise-free $\widehat{Y}$. Next, are different crops extracted from the corresponding images.

art methods in terms of PSNR and flexibility, and experiments on SBF-SEM data illustrate effective stripe removal while preserving real EM image content. Overall, the developed approach offers promising stripe noise characterization and removal, provided the noise stationarity and stripe/non-stripe content separability conditions are met.

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